

FINAL EXPLANATION: GENERAL SCIENCE GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: GENERAL SCIENCE GRADE 10 – SUB-STRAND 2.2 CHEMICAL FAMILIES: FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

You have spent eight lessons collecting evidence about chemical families. Now you will use it to answer the Driving Question: Why do some elements explode in water, some glow silently in glass tubes, and others rust on a Kenyan farm, and how do their family properties explain the materials that build, light up, and sustain our daily lives? Answer each section in the space below its prompt. Write in full sentences. Use the Claim-Evidence-Reasoning (CER) pattern where asked: make a claim, back it with evidence from our lessons, and explain your reasoning using electrons. Refer to the Thika Road street lamp, the Kamukunji welding shield and the helium balloon. Include balanced equations where the prompt asks for them.

Part 1: What is a chemical family?

Sodium, argon and helium are all elements, yet they behave very differently. (a) In your own words, what is a chemical family? (b) What single feature of an atom decides which family an element belongs to? (c) Name four families we studied in detail and give one element from each.

(a) A chemical family is a group of elements in the same column of the periodic table that have similar properties and behave in similar ways. (b) The number of valence (outer-shell) electrons decides the family, because elements in the same family have the same number of outer electrons and so behave alike. (c) Alkali metals (Group 1): sodium. Alkaline earth metals (Group 2): magnesium or calcium. Noble gases (Group 18): helium or argon. Transition metals: iron, copper or zinc.

Part 2: Alkali metals and the Thika Road street lamp (CER)

Write a Claim-Evidence-Reasoning answer to this question: Why does sodium react violently with water, while lithium reacts less vigorously and potassium more vigorously? (a) Give a claim. (b) Give evidence from our water-reaction demonstration. (c) Explain your reasoning using valence electrons and shells.

(a) Claim: Alkali metals have one loosely held outer electron, and it is lost more easily as you go down the group, so reactivity increases $\text{Li} < \text{Na} < \text{K}$. (b) Evidence: In the demonstration, lithium, sodium and potassium all fizzed and gave off hydrogen gas, and phenolphthalein turned pink showing an alkaline solution. The reaction got faster from lithium to sodium to potassium, and potassium burned with a lilac flame. (c) Reasoning: All Group 1 metals have 1 outer electron. Going down the group each atom has one more electron shell, so the outer electron is further from the nucleus and shielded by more inner shells. The attraction to the nucleus is weaker, so the electron is lost more easily and the metal reacts more vigorously. (d) $2\text{Na} + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH} + \text{H}_2$. (e) In the Thika Road lamp tube, sodium does not react chemically. Its loosely held outer electron takes in electrical energy and gives it out again as orange-yellow light. So the same loosely held outer electron is linked to both the reaction with water and the light, but the lamp involves no chemical reaction.

(d) Write the balanced equation for sodium reacting with water. (e) Sodium is also the element inside the Thika Road street lamp. What does this tell you about how sodium atoms can take in energy and give out light?	
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Part 3: Alkaline earth metals and Kenyan farms

Magnesium is in Group 2 and sodium is in Group 1. (a) Compare how magnesium and sodium reacted in our experiments. (b) Use the number of outer electrons and the energy needed to remove them to explain the difference. (c) Write the balanced equation for magnesium reacting with dilute hydrochloric acid. (d) Calcium is a Group 2 element, and CAN fertiliser used by Rift Valley maize farmers contains calcium. Explain why a Group 2 element can be useful and safe in a fertiliser bag stored in an agrovet shop, whereas sodium metal could not be.	(a) Sodium reacts violently with water. Magnesium reacted only very slowly with cold water, moderately with steam (making white magnesium oxide and hydrogen), and bubbled vigorously in dilute HCl, but all of its reactions were slower and less dramatic than those of sodium. (b) Magnesium has two outer electrons (2,8,2) and sodium has one. Both of magnesium's electrons must be lost to form Mg^{2+}, and removing two electrons takes more energy than removing one, so Group 2 metals are less reactive than Group 1. (c) $Mg + 2HCl \rightarrow MgCl_2 + H_2$. The hydrogen gas gives a squeaky pop with a lit splint. (d) Group 2 elements are only moderately reactive because losing two outer electrons needs more energy, so they are not dangerous like sodium. In CAN fertiliser, calcium is present in a compound as Ca^{2+} ions, not as the metal, and it feeds crops when spread on maize fields. This is safe to store in agrovet shops, whereas sodium metal would react violently with water.
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Part 4: Noble gases and transition metals

(a) Explain why argon is used in the Kamukunji welding shield and helium in the party balloon, using the idea of outer shells. (b) Iron in a jembe and zinc on a mabati roof are transition metals. Describe how their reactivity compares with sodium and how this affects their use. (c) Write a balanced equation for one reaction of	(a) Argon (2,8,8) and helium have full outer shells, so they have no tendency to gain, lose or share electrons. Argon therefore does not react with the hot metal or oxygen, making it a safe protective shield, and helium is inert and harmless, making it safe for balloons. (b) Transition metals react much more slowly than sodium: iron and zinc react slowly with dilute acid and copper does not, and none react explosively with water. A jembe rusts slowly over months instead of exploding. Their hardness, malleability and conductivity make iron good for tools, copper for wiring and zinc for coating mabati. (c) For example: $2Na + 2H_2O \rightarrow 2NaOH + H_2$ (sodium, an alkali metal in Group 1), or $Mg + 2HCl \rightarrow MgCl_2 + H_2$ (magnesium, an alkaline earth metal in Group 2).
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magnesium or sodium that we studied, and say which family the metal belongs to.	
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Part 5: Answering the Driving Question

Write a connected paragraph (6-8 sentences) answering the Driving Question: Why do some elements explode in water, some glow silently in glass tubes, and others rust on a Kenyan farm, and how do their family properties explain the materials that build, light up, and sustain our daily lives? Mention at least three families and one Kenyan example for each. Then state how your thinking has changed since Lesson 1.	The behaviour of an element depends on its valence electrons, which place it in a chemical family. Alkali metals such as sodium have one loosely held outer electron, so they explode in water, and sodium glows orange in Thika Road street lamps. Noble gases such as argon and helium have full outer shells, so they do not react; this makes argon a safe shield for jua kali welding in Kamukunji and helium safe for balloons. Alkaline earth metals such as calcium have two outer electrons and moderate reactivity, and calcium compounds are used in CAN fertiliser for Rift Valley maize. Transition metals such as iron, copper and zinc react slowly, so jembes, wiring and mabati roofs last a long time and rust only gradually. In Lesson 1 I could only say the elements were different; now I can predict how an element behaves from its family and its outer electrons.
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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Use of chemical family and valence electron ideas	Clearly and accurately links every family to its number of valence electrons and uses this to explain behaviour in every part.	Links most families to valence electrons correctly, with minor gaps or imprecise wording.	Names families or elements but does not connect behaviour to valence electrons.
Claim-Evidence-Reasoning structure	Gives a clear claim, specific evidence from the lessons (observations such as fizzing, pink phenolphthalein, squeaky pop) and reasoning that links the evidence to electron arrangement.	Gives a claim and some evidence, but the reasoning is partly missing or not clearly linked.	Gives statements without evidence or reasoning.
Balanced equations and chemical accuracy	All required equations are correct and balanced, and the products are correctly identified.	Equations are mostly correct with one small error in balancing or state.	Equations are missing or incorrect.
Connection to the anchoring phenomenon and Kenyan contexts	Explains the Thika Road lamp, Kamukunji welding and helium balloon, plus farm and hardware examples, using family properties.	Explains most of the phenomenon examples with some Kenyan contexts.	Mentions few examples or does not link them to chemical families.

Clarity and growth in understanding	Writes a well-organised, connected paragraph and clearly shows how their ideas changed since Lesson 1.	Writes a mostly clear explanation with a brief note on how their ideas changed.	Writing is disconnected and does not show how their ideas changed.
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ATTRIBUTION AND LICENSING

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Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

Production & Layout: Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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